

Lecture 13

Objective priors: Part 2

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(Edited from the lecture notes by Brian J. Reich, NC State)



Prior elicitation: Recap

- ▶ Potential sources of prior information include: literature reviews; pilot studies; expert opinions; etc.
- ▶ Prior elicitation is the process of converting expert opinion to prior distribution.
- ▶ Prior information can come from a variety of sources and the mixture of experts prior is

$$\pi(\theta) = \sum_{j=1}^J w_j \pi_j(\theta).$$

- ▶ Another powerful idea is to elicit prior information using real-life scenarios.

Improper priors and objective priors: Recap

- ▶ A ‘prior’ that integrates to infinity is called an improper prior.
- ▶ Example: $\pi(\mu) = 1$ for all $\mu \in \mathbb{R}$.
- ▶ It’s OK to use an improper prior as long as you verify that the posterior integrates to one.
- ▶ An objective analysis is one that requires no subjective decisions by the analyst.
- ▶ There are many approaches: Jeffreys, reference, probability matching, maximum entropy, empirical Bayes, penalized complexity, etc.

Jeffreys prior: Recap

- ▶ The Jeffreys prior (JP) for parameter θ is $p(\theta) \propto \sqrt{I(\theta)}$.
- ▶ The Fisher information is $I(\theta) = -\mathbb{E}_{Y|\theta} \left[\frac{d^2}{d\theta^2} \log p(Y|\theta) \right]$.
- ▶ Once you have specified the likelihood, the Jeffreys prior is determined with no additional input.

Likelihood	Jeffreys prior
▶ $Y \sim \text{Binomial}(n, \theta)$	$\theta \sim \text{Beta}(1/2, 1/2)$
$Y \sim \text{N}(\mu, 1)$	$p(\mu) \propto 1$
$Y \sim \text{N}(0, \sigma^2)$	$p(\sigma) \propto 1/\sigma$

Why Jeffreys prior?

- ▶ You use a JP on θ , denoted $\pi_1(\theta)$, and I place a JP on $\gamma = g(\theta)$, denoted $\pi_2(\gamma)$.

- ▶ Using the change of variables formula, the prior for γ induced by $\pi_1(\cdot)$ is

$$\pi_3(\gamma) = \pi_1(g^{-1}(\gamma)) \left| \frac{dg^{-1}(\gamma)}{d\gamma} \right|.$$

- ▶ It can be shown that $\pi_2(\cdot) = \pi_3(\cdot)$.
- ▶ This establishes invariance to reparameterization.

An example

- ▶ Likelihood: $Y \sim \text{Binomial}(n, \theta)$. Here, JP is $\theta \sim \text{Beta}(1/2, 1/2)$.
- ▶ You choose θ as your parameter and thus $\pi_1(\theta) = \frac{1}{\pi} \theta^{1/2-1} (1 - \theta)^{1/2-1}$.
- ▶ I choose $\gamma = \theta^2 \in [0, 1]$ as my parameter. Thus, $g(x) = x^2$.
- ▶ The prior for γ induced by $\pi_1(\cdot)$ is

$$\pi_3(\gamma) = \frac{1}{2\pi} (\sqrt{\gamma})^{-1} (1 - \sqrt{\gamma})^{1/2-1}.$$

- ▶ Now, I rewrite the model as $Y \sim \text{Binomial}(n, \sqrt{\gamma})$. Then obtain $\pi_2(\cdot)$.
- ▶ After a few steps of boring algebra, we get $\pi_2(\cdot) = \pi_3(\cdot)$.

Reference priors

- ▶ These priors try to formally be ‘non-informative’.
- ▶ Similar to JP, they are objective.
- ▶ They are defined as the prior that gives the maximum expected ‘distance’ between the log-prior and log-posterior (Kullback-Leibler divergence).
- ▶ For univariate models, they give JP. For multivariate models, they are different.
- ▶ Reference priors are harder to compute than JP.
- ▶ The solution of the optimization problem is a PDF and not a scalar or vector.

Probability matching priors (PMP)

- ▶ The PMP is designed so that posterior credible intervals have correct frequentist coverage.
- ▶ There are only a few cases where this can be done exactly.
- ▶ For example, if $Y_1, \dots, Y_n | \mu \sim \text{Normal}(\mu, 1)$, the PMP is $p(\mu) = 1$.
- ▶ The posterior is $\mu | Y_1, \dots, Y_n \sim \text{Normal}(\bar{Y}, 1/n)$.
- ▶ In this case, credible sets have correct frequentist coverage.
- ▶ Approximations are available for medium-complexity cases.

Empirical Bayes

- ▶ Empirical Bayes is also objective.
- ▶ Here you pick the priors based on the data.
- ▶ Example: σ^2 has prior mean s^2 .
- ▶ Example: σ^2 is fixed at s^2 in the Bayesian analysis of μ .
- ▶ More formally, you can use marginal maximum likelihood to fix nuisance parameters. The analysis then proceeds as a usual Bayesian analysis.
- ▶ Often criticized for 'using the data twice'.
- ▶ Despite this concern, EB remains a useful tool to stabilize computing in high-dimensional problems.

Penalized complexity (PC) priors

- ▶ A PC prior begins with a simple base model, e.g., linear regression with all the slopes equal zero.
- ▶ The full model, e.g., regression with non-zero slopes, is shrunk towards to the base model.
- ▶ The 'distance' of the full model to the base model has exponential prior to penalize the more complex model from deviating from the base model.
- ▶ Requires picking the parameter in the exponential prior and setting priors for the parameters in the base model.
- ▶ So technically this is not purely objective.

Maximum entropy priors

- ▶ In a maximum entropy prior you fix a few quantities of the prior distribution, e.g., $E(\theta) = 0.5$.
- ▶ The prior is taken to be the distribution with the maximum entropy of these that satisfy the known constraints.
- ▶ **Entropy** is a measure of uncertainty, e.g., the entropy of the PMF $f(x)$ is

$$-\sum_{x \in \mathcal{S}} f(x) \log[f(x)]$$

- ▶ If θ has support $\mathbb{R}$ and the mean and variance are known, the maximum entropy prior is Gaussian.
- ▶ Not purely objective because you have to set the constraints.

Problem discussion

Improper prior example 1

Say $Y|\mu \sim \text{Normal}(0, 1)$ and μ has improper prior $\pi(\mu) = 1$ for all μ . Prove that the posterior distribution of μ is a proper PDF.

- ▶ The posterior is

$$\begin{aligned} p(\mu|Y) &\propto f(Y|\mu) \times \pi(\mu) \\ &\propto \exp\left[-\frac{1}{2}(Y - \mu)^2\right] \\ &\propto \exp\left[-\frac{1}{2}(\mu - Y)^2\right]. \end{aligned}$$

- ▶ By comparing the kernel of the posterior with that of a normal density, we have

$$\mu|Y \sim \text{Normal}(Y, 1).$$

Improper prior example 2

Suppose that the likelihood $f(Y|\theta) \geq c(Y)$ for all θ . Show that an improper prior on θ will yield an improper posterior.

- ▶ For some positive constant $c_0(Y)$ (dependent on Y but not on θ), the posterior is

$$\begin{aligned} p(\theta|Y) &= c_0(Y) \times f(Y|\theta) \times \pi(\theta) \\ &\geq c_0(Y) \times c(Y) \times \pi(\theta). \end{aligned}$$

- ▶ Thus, we have

$$\begin{aligned} \int p(\theta|Y) d\theta &\geq c_0(Y) \times f(Y|\theta) \times \int \pi(\theta) d\theta \\ &\geq c_0(Y) \times c(Y) \times \infty \\ &= \infty. \end{aligned}$$

Improper prior example 3

Consider a sample of size n from the mixture of two normal densities given by $f(y|\theta) = 0.5\phi(y) + 0.5\phi(y - \theta)$, where $\phi(z)$ denotes the standard normal density function. Show that any improper prior for leads to an improper posterior.

- For some positive $c_0(\mathbf{Y})$ (dependent on $\mathbf{Y}$ but not on θ), the posterior is

$$\begin{aligned} p(\theta|\mathbf{Y}) &= c_0(\mathbf{Y}) \times f(\mathbf{Y}|\theta) \times \pi(\theta) \\ &= c_0(\mathbf{Y}) \times \prod_{i=1}^n \{0.5\phi(y_i) + 0.5\phi(y_i - \theta)\} \times \pi(\theta) \\ &\geq c_0(\mathbf{Y}) \times \prod_{i=1}^n \{0.5\phi(y_i)\} \times \pi(\theta) = d_0(\mathbf{Y}) \times \pi(\theta). \end{aligned}$$

- Thus, $\int p(\theta|\mathbf{Y})d\theta \geq d_0(\mathbf{Y}) \times \int \pi(\theta)d\theta = d_0(\mathbf{Y}) \times \infty = \infty$.

Improper prior example 4

Say $Y|\lambda \sim \text{Poisson}(\lambda)$. (a) Derive the Jeffreys' prior for λ . (b) Is this prior proper? (c) Derive the posterior and give conditions on Y to ensure it is proper.

- ▶ The log likelihood is

$$\ell(\lambda) = \log[f(Y|\lambda)] = -\lambda + Y \log(\lambda) - \log(Y!).$$

- ▶ The first derivative is

$$\ell'(\lambda) = -1 + Y/\lambda.$$

- ▶ The second derivative is

$$\ell''(\lambda) = -Y/\lambda^2.$$

- ▶ Therefore,

$$I(\lambda) = -E(\ell''(\lambda)) = \lambda/\lambda^2 = 1/\lambda.$$

- ▶ Finally, the prior is

$$\pi(\lambda) \propto I(\lambda)^{1/2} = \lambda^{-1/2}.$$

Improper prior example 4 (Contd.)

- ▶ $\int \pi(\lambda) d\lambda \propto \int_0^\infty \lambda^{-1/2} d\lambda = \infty$. Thus, improper prior.
- ▶ The posterior is $p(\lambda|Y) \propto \exp[-\lambda] \lambda^Y \times \lambda^{-1/2} = \exp[-\lambda] \lambda^{Y+1/2-1}$.
- ▶ Thus, $\lambda|Y \sim \text{Gamma}(Y + 1/2, 1)$.
- ▶ Thus, the posterior is proper for all Y .

Improper prior example 5

Say $Y|\lambda \sim \text{Exponential}(\lambda)$. (a) Derive the Jeffreys' prior for λ . (b) Is this prior proper? (c) Derive the posterior and give conditions on Y to ensure it is proper.

- ▶ The log likelihood is

$$\ell(\lambda) = \log[f(Y|\lambda)] = \log(\lambda) - \lambda Y.$$

- ▶ The first derivative is

$$\ell'(\lambda) = 1/\lambda - Y.$$

- ▶ The second derivative is

$$\ell''(\lambda) = -1/\lambda^2.$$

- ▶ Therefore,

$$I(\lambda) = -E(\ell''(\lambda)) = 1/\lambda^2.$$

- ▶ Finally, the prior is

$$\pi(\lambda) \propto I(\lambda)^{1/2} = \lambda^{-1}.$$

Improper prior example 4 (Contd.)

- ▶ $\int \pi(\lambda) d\lambda \propto \int_0^\infty \lambda^{-1} d\lambda = \infty$. Thus, improper prior.
- ▶ The posterior is $p(\lambda|Y) \propto \lambda \exp[-\lambda Y] \times \lambda^{-1} = \exp[-\lambda Y]$.
- ▶ Thus, $\lambda|Y \sim \text{Exponential}(Y)$.
- ▶ Thus, the posterior is proper for all Y .

Thank you!